

Research and Applications

Disparities in seizure outcomes revealed by large language models

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Abstract

Objective: Large-language models (LLMs) can potentially revolutionize health care delivery and research, but risk propagating existing biases or introducing new ones. In epilepsy, social determinants of health are associated with disparities in care access, but their impact on seizure outcomes among those with access remains unclear. Here we (1) evaluated our validated, epilepsy-specific LLM for intrinsic bias, and (2) used LLM-extracted seizure outcomes to determine if different demographic groups have different seizure outcomes.

Materials and Methods: We tested our LLM for differences and equivalences in prediction accuracy and confidence across demographic groups defined by race, ethnicity, sex, income, and health insurance, using manually annotated notes. Next, we used LLM-classified seizure freedom at each office visit to test for demographic outcome disparities, using univariable and multivariable analyses.

Results: We analyzed 84 675 clinic visits from 25 612 unique patients seen at our epilepsy center. We found little evidence of bias in the prediction accuracy or confidence of outcome classifications across demographic groups. Multivariable analysis indicated worse seizure outcomes for female patients (OR 1.33, $P \leq .001$), those with public insurance (OR 1.53, $P \leq .001$), and those from lower-income zip codes (OR ≥ 1.22 , $P \leq .007$). Black patients had worse outcomes than White patients in univariable but not multivariable analysis (OR 1.03, $P = .66$).

Conclusion: We found little evidence that our LLM was intrinsically biased against any demographic group. Seizure freedom extracted by LLM revealed disparities in seizure outcomes across several demographic groups. These findings quantify the critical need to reduce disparities in the care of people with epilepsy.

Key words: electronic health record; natural language processing; clinical informatics; health disparities.

Introduction

In the US healthcare system, minoritized and marginalized groups experience profound disparities in access to care and health outcomes.¹ In neurology, for example, non-White patients have less access to neurologic care than White patients and worse health outcomes across many neurologic conditions and subspecialties.^{2,3} Epilepsy is a common neurologic disorder characterized by recurrent, unprovoked seizures. For people with epilepsy, demographic factors play a role in access to high-quality medical care.^{4–7} For example, studies in the United States have found that race and socioeconomic status are associated with access to neurologists and epilepsy specialists,^{8,9} access to appropriate diagnostic testing,⁹ and referral for surgical treatment when it is indicated for medication-resistant epilepsy.¹⁰

Most studies of disparities in epilepsy have focused on access to care.⁷ In contrast, much less is known about demographic disparities in clinical outcomes. One of the most important outcome measures for people with epilepsy is seizure freedom.^{11,12} A 2009 systematic review on disparities in epilepsy by a commission of the International League Against Epilepsy identified no studies that examined disparities in medical outcomes.⁵ Since then, 2 studies of children with epilepsy have begun to address this question. One study of 776 children in Utah found that Hispanic ethnicity was associated with time to seizure remission.¹³ A study of 1038 children in Philadelphia found that seizure freedom was associated with household income but not with race or ethnicity.¹⁴ A more complete understanding of disparities in patient outcomes is important because solutions targeting access, while necessary, will not lead to

health equity if patients have different outcomes even when they have access to specialist care.

Just as disparities exist in healthcare, there are well-founded concerns that artificial intelligence (AI) tools used to deliver or evaluate healthcare will propagate existing biases or introduce new ones.^{15–17} For example, a widely used commercial risk prediction tool to guide allocation of care resources was found to underestimate the severity of illness for Black patients relative to White patients, exacerbating a disparity in access to care.¹⁸ One form of AI, natural language processing (NLP), uses machines to read and understand human language, which enables rapid, large-scale extraction and analysis of unstructured clinical data from electronic health records. NLP is vulnerable to perpetuated bias because biases present in unstructured note text and in healthcare delivery more broadly will be incorporated into the NLP algorithm. For example, a recent algorithm for identifying opioid misuse using clinical notes was found to have bias in Black compared to White individuals.^{19,20} In the field of epilepsy, an NLP tool to identify surgical candidates was found to reflect biases present in patient referrals, such as favoring patients with higher household income.²¹ Recent developments in ethical AI have emphasized the importance of ensuring equality in model performance between protected groups.^{15,17,18,22,23}

We have recently developed and validated an NLP algorithm to extract epilepsy outcome measures from the unstructured text of clinical notes with accuracy similar to human readers.^{24–26} In this study, we first tested our algorithm for bias, in the form of differential performance in different demographic groups. An unbiased NLP algorithm was a prerequisite to our second objective, which was to use the algorithm to test the hypothesis that different demographic groups have different likelihoods of seizure freedom at our academic US healthcare center.

Methods

This research was approved by the Institutional Review Board of the University of Pennsylvania.

Data collection

Our source dataset consisted of electronic health records from patients who had seen an epileptologist at the University of Pennsylvania Health System between the years of 2005 and 2022. From these records, we obtained clinical notes written by their epileptologist(s) with full text and date; medication prescriptions; and demographic information including race, sex, zip code, age, and insurance provider at each visit. We excluded patients with missing demographic data.

Outcomes: seizure classification by NLP

We recently developed and validated an NLP algorithm that classifies clinic notes as seizure-free or having recent seizures.^{24,25} Briefly, we used Clinical_BERT,²⁷ a publicly-available transformer language model from Google AI, fine-tuned on 700 manually annotated epileptologist notes. We defined a “seizure free” visit as one where the patient did not have seizures since their last visit, or within the past year, whichever was more recent. Model predictions were repeated 5 times using different seeds, and final classification of each note was determined by plurality voting of the 5 outputs.

Exposures: demographic variables

Race, ethnicity, and sex were self-reported by patients at the time of initial registration with our health system, while zip code and insurance were entered and verified by clinic registration staff. Category options were determined by the Epic electronic health record software (Epic, Madison WI). Race category options were as follows, listed alphabetically: American Indian or Alaskan Native; Asian; Black or African American; East Indian; Native Hawaiian or Other Pacific Islander; Some Other Race; White. Only Asian, Black, and White had sufficient samples for analysis, and the other categories were combined into a category of “Other Races.” Ethnicity categories were Hispanic Latino or Not Hispanic Latino. Sex categories were female or male. Gender identities, including trans and non-binary genders, were not recorded.

Zip codes were linked to median household income, in inflation-adjusted 2020 dollars, using publicly available data from the US Census (data.census.gov). We grouped zip codes into 4 categories according to median household income: (1) less than \$50 000; (2) \$50 000 to less than \$75 000; (3) \$75 000 to less than \$100 000; (4) \$100 000 or more. Insurance type was classified as private insurance, Medicare, or Medicaid. For purposes of analysis we combined Medicare and Medicaid into a single category of public insurance. Age was grouped into the following categories: 18–39, 40–64, and 65 or older.

Assessing model bias

These analyses used 192 manually-annotated notes from the validation dataset from our previous study.²⁴ Human annotations were performed in triplicate by independent readers who were unaware of the identities or demographics of the patients. Annotations were merged by majority voting with manual adjudication to resolve disagreements.²⁴ Here we sought to determine whether our NLP models performed differently in different demographic groups using several measures. First, we calculated the accuracy of the model classifications (seizure free versus recent seizure), where accuracy was the number of correct classifications divided by all classifications (Supplementary methods).²⁴ Second, we calculated the positive class balance (PCB) and negative class balance (NCB) of each demographic group in order to evaluate over and under-reporting of outcomes.^{23,28} These methods use the confidence values that accompany each model prediction (eg, the model may predict that a note describes recent seizures with confidence of 0.92). PCB is the average predicted confidence of recent seizures in patients who did have recent seizures, with an expected value near 1. NCB is the average predicted confidence of recent seizures in patients who did not have recent seizures, with an expected value near 0. If the NLP model made predictions with perfect confidence, the PCB would be the complement of the false negative rate ($FNR = 1 - PCB$) and the NCB would be the equivalent of the false positive rate ($FPR = NCB$).

Statistical analysis

We evaluated model bias by comparing the model’s accuracy between demographic groups using two strategies: testing for equivalence and testing for difference. We first dichotomized each demographic group to maximize statistical power, as follows: male or female sex; Hispanic/Latino or not Hispanic/Latino ethnicity; White or non-White race; above or below the median income of \$82 158; above or below the median

age of 46; private or public insurance. Although combining non-White races may be considered pejorative,²⁹ we faced a tradeoff with statistical power and wanted to maximize our ability to detect model bias. We were unable to test for significant equivalences or differences in model accuracy for ethnicity due to the small number of Hispanic/Latino subjects among manually annotated notes ($n = 7$).

To test for equivalence, for each demographic variable, we compared model accuracy in the 2 dichotomous groups using the test of 2 proportions to generate an effect size estimate (proportion difference) and confidence interval. We then performed equivalence testing using the method of two one-sided tests (TOST),³⁰ with a 90% confidence interval around the effect size estimate and an equivalence margin of 10%.^{31,32}

To test for difference, we used Fisher’s exact tests to compare model accuracy between the dichotomized demographic groups within each demographic variable. To test for differences in PCB and NCB, we first calculated the logistic transformation of the model confidence values. For each demographic variable, we then performed linear regression to test for differences in the mean logit-transformed PCB and NCB values between demographic groups. We did not perform equivalence testing for PCB and NCB because of insufficient power: because PCB and NCB are restricted to true positives and true negatives, respectively, there were fewer observations than for accuracy.

We considered a lack of significant difference to be sufficient evidence of a lack of model bias to proceed with the disparities analysis for each demographic variable. Though other studies have used calibration to assess model bias,³³ we instead chose to use class balances to ensure that our model was not over or under predicting seizure freedom for specific demographic groups; as calibration and class balances are mutually exclusive metrics, we did not compare the calibration of our model.²⁸

Next, we tested for disparities in seizure outcomes in the entire source dataset using NLP-derived outcomes. We performed a series of logistic mixed effects regression models to identify associations between demographics and seizure freedom classification. In each model the outcome was the seizure freedom classification of each visit; exposure was the demographic variable; and patient was the clustering variable (random effect) to account for intra-individual correlation across repeated measures. We included the time (in months) since last visit as a covariate in each model, to account for the fact that patients with frequent seizures are likely to be seen more often than seizure-free patients.²⁶ Additionally, we included the number of anti-seizure medications (ASMs) a patient was prescribed at the time of a visit as a variable of interest to act as a positive control, as we expected to find an association between seizure freedom likelihood and number of ASMs.

First we tested each demographic variable in a separate mixed effects model, representing “univariable” analyses, adjusting only for time since last visit, and patient-specific random effect. Second, we combined all demographic variables in a single mixed-effects model, representing a multivariable analysis to assess the competing effects of demographic variables. We examined our logistic regression and found that they did not violate the basic assumptions for such models (Supplementary results).

Table 1. Summary of patient-reported demographic information of our patient cohort.

Demographic	N (%) of 25 612 patients
Race	
American Indian or Alaskan Native	67 (0.3%)
Asian	741 (2.9%)
Black or African American	8308 (32.4%)
East Indian	6 (0.0%)
Native Hawaiian or Other Pacific Islander	15 (0.1%)
Some Other Race	990 (3.9%)
White	13 916 (54.3%)
Ethnicity	
Hispanic Latino	826 (3.2%)
Not Hispanic or Latino	24 313 (94.9%)
Sex	
Male	11 078 (43.3%)
Female	14 534 (56.7%)
Age at latest visit	
18-39	7785 (30.4%)
40-64	10 081 (39.4%)
65 or older	7746 (30.2%)
Insurance	
Public	14 830 (57.9%)
Private	9227 (36.0%)
Median income per zip code	
Less than \$50 000	6960 (27.2%)
\$50 000 to <\$75 000	5646 (22.0%)
\$75 000 to <\$100 000	6452 (25.2%)
Over \$100 000	6384 (24.9%)

We adjusted *P*-values for multiple comparisons in both the univariable regression and bias equivalence analyses using the Benjamini-Hochberg false discovery rate method with an α of .05.³⁴ We did not correct for multiple comparisons when testing for differences because this correction would make it harder to detect evidence of model bias.

Results
Cohort

Demographic information was available for 25 612 patients and is summarized in Table 1. Of the 84 675 visits from which seizure freedom could be determined, 22 038 (26%) were classified as seizure free and 48 327 (57%) were classified as having recent seizures. This distribution of seizure free, not seizure free, and unclassifiable notes is similar to what has been previously seen from the health records from our center from automated and manual review.^{24,26} A total of 3265 patients were excluded due to missing or incomplete demographics and 14 310 visits were excluded due to unclassifiable seizure freedom. To visualize the spatial distribution of our patient cohort in our local tri-state area, we have also made interactive maps of our patients and some of their demographic variables, by zip code (Supplementary results).

No evidence of model bias

The accuracy of the NLP model did not differ across all demographic variables (Fisher’s exact $P \geq .149$). Accuracy was also statistically equivalent between subgroups of sex, age, and income (TSTP $P \leq .049$), and inconclusive between subgroups of insurance and race (Figure 1). After correcting for multiple comparisons, no subgroups had significantly equivalent accuracies. NCB did not differ across all demographic

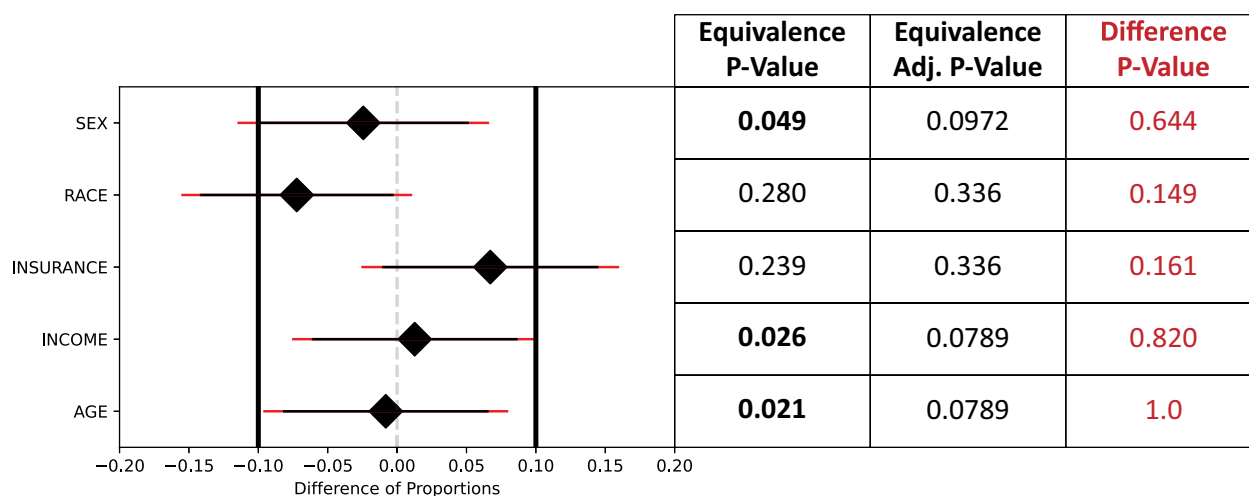


Figure 1. Bias in model accuracy. Equivalence and difference testing for model bias across binarized demographic variables. The forest plot shows the difference of proportions of correct predictions between the 2 demographic groups and associated 90% (black) and 95% (red) confidence intervals. The table on the right shows the associated *P*-values. Equivalence testing used 2 one-sided tests of equivalence against a 90% CI with an equivalence threshold of 0.10 (vertical black bars). Difference testing used a proportion test of differences against a 95% CI for each demographic variable. Bolding indicates significance.

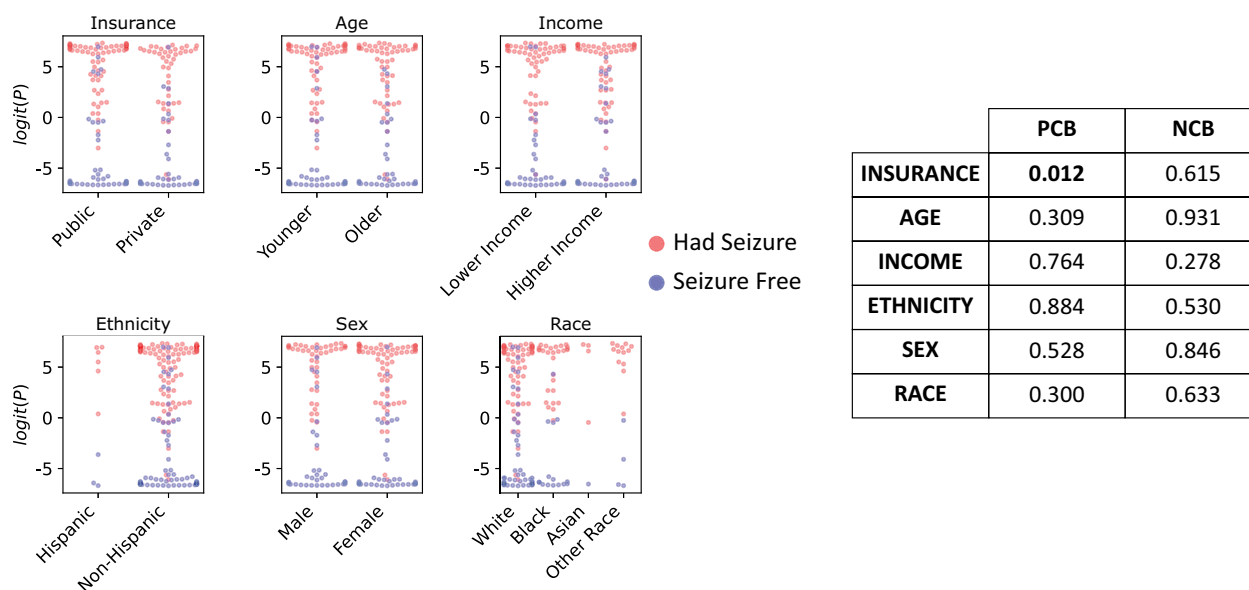


Figure 2. Bias in class balance. Difference testing for class balance between demographic groups. Plots on the left show the distributions of the logit transformed model confidence values for true positives (red) and true negatives (blue) between different demographic groups. The table contains the *P*-values for the binomial regression model used to compare the predicted probabilities of true positives (PCB) and true negatives (NCB) between demographic groups. Bolding indicates significance.

variables (regression test $P \geq .278$). PCB was significantly different only between subgroups of insurance (regression test $P = .012$) (Figure 2). Because we did not see significant differences for all but one of our demographic groups, we felt comfortable proceeding with our disparities analyses while being conservative when interpreting differences based on insurance.

Disparities in seizure freedom classification

Disparities in the likelihood of seizure-free visits were present for each demographic variable we analyzed, with the minoritized and marginalized groups having worse outcomes than privileged groups (Figure 3). Specially, female patients (OR 1.35, 95% CI 1.21-1.50, adj. $P < .001$), Black patients (OR

1.40, 95% CI 1.24-1.58, adj. $P < .001$), patients with public insurance (OR 1.48, 95% CI 1.33-1.64, adj. $P < .001$), and patients living in zip codes with less than \$100 000 median income (OR ≥ 1.17 , adj. $P \leq .037$), were more likely to have recent seizures than patients from privileged groups. Older patients were more likely to be seizure free than younger patients. Greater numbers of prescribed ASMs were associated with lower likelihood of seizure freedom, as expected.

Some but not all of these disparities persisted in the multi-variable regression analysis (Table 2). Black patients no longer had significantly higher odds of seizures, and patients of Other Races had significantly lower odds of seizures than White patients, indicating some overlap between the effects of race and other demographic variables.

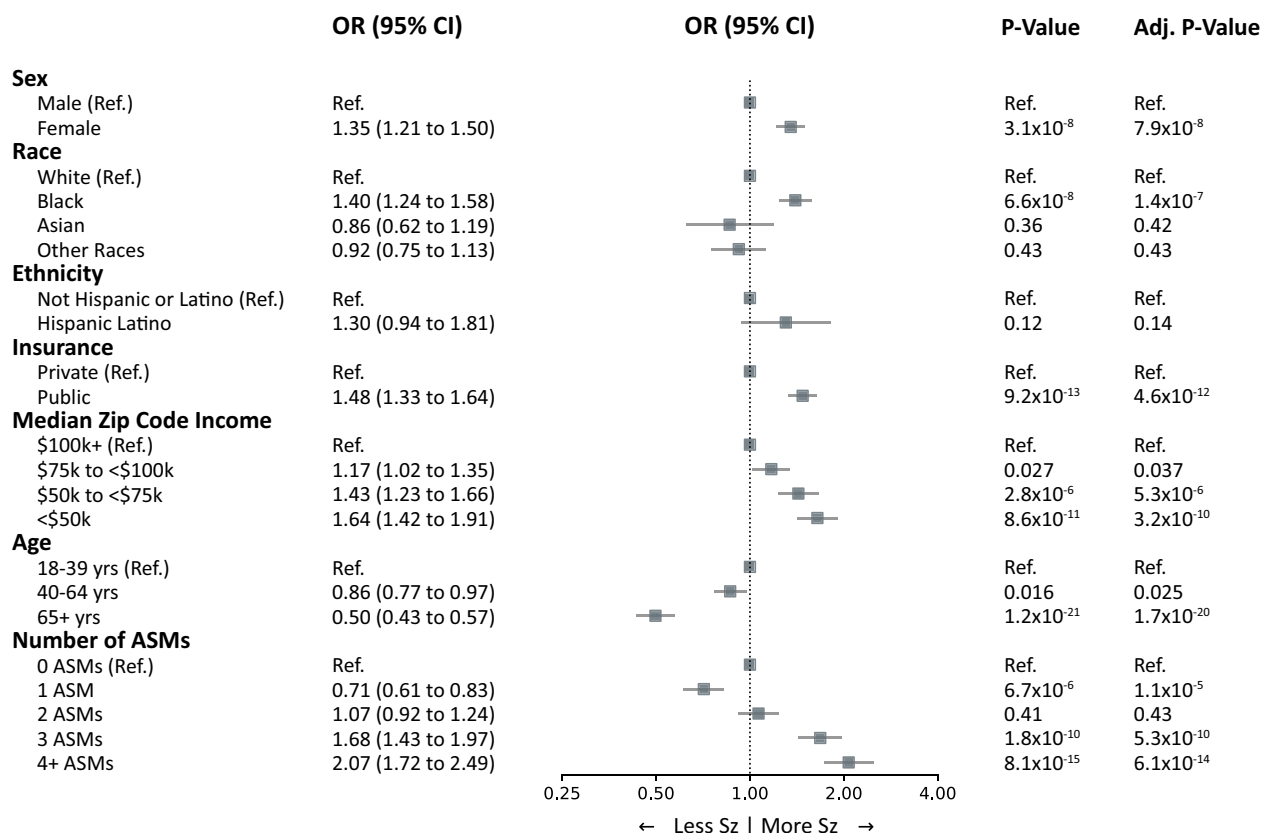


Figure 3. Disparities in seizure freedom likelihood. Forest plot with univariable estimates of odds ratios (OR) of recent seizures at each office visit. ORs were estimated from logistic mixed model regression at the visit level, grouping visits by patients, and controlling for time since last visit. "Other Races" included "American Indian or Alaskan Native," "East Indian," "Native Hawaiian or Other Pacific Islander," and "Some Other Race." "Anti-Seizure Medications" were abbreviated as ASMs. Adjusted *P*-values were calculated using the Benjamini-Hochberg false discovery rate correction with an α .05.

Table 2. Multivariable analysis of demographic variables and seizure freedom.

Variable	Estimate	OR	95% CI	Multivariate P-value	Univariate Adj. P-value
Sex (Ref. Men)	0.28	1.33	1.20-1.46	<.001	<.001
Race (Ref. White)					
Black	0.03	1.03	0.89-1.19	.66	<.001
Asian	-0.21	0.81	0.60-1.09	.17	.42
Other races	-0.22	0.80	0.65-0.98	.035	.43
Ethnicity (Ref. Not Hispanic or Latino)	0.15	1.16	0.83-1.61	.38	.14
Insurance type (Ref. Private)	0.42	1.53	1.37-1.71	<.001	<.001
Median zip code income (Ref. \$100k+)					
\$75k to <\$100k	0.11	1.11	0.97-1.27	.11	.037
\$50k to <\$75k	0.20	1.22	1.06-1.41	.0066	<.001
<\$50k	0.28	1.33	1.11-1.58	.0014	<.001
Age (Ref. 18-39 years)					
40-64 years	-0.23	0.80	0.71-0.89	<.001	.025
65+ years	-0.84	0.43	0.37-0.50	<.001	<.001
Number of anti-seizure medications taken (Ref. 0 ASMs)					
1 ASM	-0.33	0.72	0.62-0.83	<.001	<.001
2 ASMs	0.05	1.05	0.91-1.23	.49	.43
3 ASMs	0.49	1.64	1.39-1.92	<.001	<.001
4 ASMs	0.69	1.99	1.65-2.39	<.001	<.001
Months since last visit	-0.02	0.98	0.98-0.99	<.001	N/A

Discussion

In this study, using NLP-derived seizure freedom data from over 25 000 patients with epilepsy across nearly 85 000 clinic visits, we found disparities in the likelihood of seizure freedom according to demographic factors. Female sex, Black

race, public insurance, and lower income zip-codes were all associated with higher likelihood of seizures at each clinic visit, controlling for the time between visits. The disparities by sex, insurance, and income persisted after controlling for other demographic factors.

We found generally no significant differences in PCB, NCB, and accuracy. While proving equivalence is difficult, in our attempt to be rigorous despite small sample sizes we found statistically marginal evidence of equivalence only without correction. The model did not perform differently across almost all demographic groups, with the only exception being PCB for insurance. Our model's significant underreporting of seizures in privately insured individuals may affect our findings that publicly insured individuals are more likely to have poorly managed epilepsy. Our presentation and analysis of model bias aligns with the American Medical Association's policy recommendations that emphasized the development of health care AI with a mind to identify biases and prevent exacerbation of health disparities.^{16,35} Our finding is particularly important because other pretrained transformer models have exhibited bias in several contexts.^{36,37} Most relevant to our study, Wissel et al. trained a support vector machine classifier to identify potential surgical candidates among patients with epilepsy.²¹ The algorithm gave higher scores to patients with higher family income and public insurance, among other social and demographic factors, reflecting the biases present in patterns of patient referrals. The observation that our model was generally unbiased suggests that disparities in seizure freedom likelihood were not influenced by model failures.

Our findings indicate that seizure freedom likelihood is influenced by demographic factors. Prior studies of disparities in the care of patients with epilepsy have focused mainly on access to specialty care,^{4,6,7,38} such as access to neurologists and epilepsy specialists,^{8,9} diagnostic testing,⁹ medication adherence,^{39–43} and surgical referrals.¹⁰ Despite the concern that unequal access will lead to unequal outcomes, there has been relatively little study of disparities in seizure outcomes. One study from Utah measured time to seizure remission in 776 children with epilepsy.¹³ The cohort was 73% White, 14% Hispanic, 5% Other Races, and the remaining 8% unspecified. In univariate analyses, factors associated with seizure remission were age, area deprivation index, and years of follow up, but not sex, race/ethnicity, or insurance. In multivariable Cox proportional hazards regression analyses, Hispanic ethnicity and public insurance were associated with longer time to remission, despite similar rates of MRIs, EEGs, and inpatient days. A second study of 1038 children in Philadelphia found that seizure freedom was associated with household income but not with race or ethnicity.¹⁴ Worsening of seizure frequency over time was associated with Hispanic ethnicity and lower household income. Third, a study of outcomes after temporal lobe epilepsy surgery in 252 patients found no significant effects of any of the predictors they examined, including race and socioeconomic status, although the study may have been underpowered.⁴⁴

Comparing our study to these prior studies, ours is the first study to analyze seizure freedom, the primary outcome of interest in the treatment of epilepsy, in a large cohort of adult patients over time. In contrast to the two pediatric studies, we found significant effects of age, sex, income, and insurance, but not of race or ethnicity after adjusting for other factors in multivariable analysis. Our study differed from these prior studies in several ways, including a cohort of adults versus children; a more diverse population; and a more nuanced accounting for the clinical reality that seizure outcome is not a fixed binary property of each patient but can change from

visit to visit.²⁶ We addressed this complexity through mixed effects models, not used by previous studies. Our findings contribute to the growing awareness that health outcomes for people with epilepsy vary across demographic factors, highlighting the critical need to understand and remedy these disparities. Our patients were all receiving care at a specialized academic epilepsy center, so the outcome disparities we observed were not due to unequal access to neurologic care per se. This suggests that structural barriers and disparities in access to healthcare resources or care delivery persist, and impact health outcomes, even in patients with subspecialist access.

It is also notable that our study was 25-times larger than the previous largest study.¹⁴ Prior to our study, a barrier to the study of seizure outcomes has been the absence of large datasets. Seizure outcomes are typically recorded only in the text of clinical notes and not easily extracted or inferred from discrete data in the electronic health record. Of the previous studies cited above, two used manual chart review and one used a limited sample of common data elements entered by clinicians. Our NLP approach allowed us to analyze nearly 25 000 patients seen at our center over 17 years, demonstrating the potential of our approach to answer important clinical research questions at large scale.

In our multivariable analysis, disparities according to sex, insurance, and income persisted after controlling for other demographic factors, while race was no longer significantly associated with the likelihood of seizure freedom. Race and socioeconomic status are strongly correlated in US society due to past and present structural racism, and this correlation means they may not be independent predictors of health outcomes. In our study, socioeconomic factors (income and insurance) were the stronger predictors of seizure freedom. Notably, the multivariable model also accounted for the number of prescribed medications and the time between clinic visits, meaning that 2 patients seen at similar intervals, prescribed the same number of ASMs, have different seizure freedom likelihoods based in part on socioeconomic factors. Previous studies of seizure outcomes did not adjust for these potential confounds.^{13,14}

Our study had several limitations. Our data do not reveal the cause of the disparities we observed. Future studies must seek to understand the causes and, more importantly, attempt interventions to reduce the disparities that are now widely recognized throughout healthcare. Our demographic categories were limited by the variables in our electronic health record, which used outdated frameworks for race and gender, and did not capture the full diversity of our patient population. Our study was performed at a United States academic medical center with a presumed bias towards more difficult and complex epilepsies. Some possible limitations of our analysis of NLP model bias include potential biases that were present in the human annotations used for both training and testing our NLP model that could be internalized by the model and not detected in its performance on the testing set. However, our human annotations were performed in triplicate by independent readers who were unaware of the identities or demographics of the patients, so bias at that step is unlikely. Additionally, analysis of PCB and NCB both reduced the size of our already limited bias validation data, reducing power in our statistical tests of difference, and precluding any statistical tests of equivalence. Future work

should further evaluate model biases using expanded validation datasets.

Conclusions

In conclusion, seizure outcomes extracted by natural language processing revealed disparities in the likelihood of seizure freedom across many social determinants of health. We found no evidence of intrinsic model biases within the NLP algorithm. These findings highlight the critical need to reduce disparities across healthcare.

Author contributions

Kevin Xie, William K.S. Ojemann, Ryan S. Gallaher, Dan Roth, Brian Litt, Colin A. Ellis conceptualized and implemented the study. Kevin Xie, William K.S. Ojemann, Ryan S. Gallaher, and Colin A. Ellis had access to and have verified the data reported in the manuscript. Russell T. Shinohara provided statistical expertise. Alfredo Lucas, Chloé E. Hill, Roy H. Hamilton, and Kevin B. Johnson provided feedback on the methods, design, and manuscript of the study. All authors were involved in the drafting, editing, and approval of the final manuscript.

Supplementary material

[Supplementary material](#) is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

None to declare.

Data availability

All analyses were performed with Python and used the following packages: transformers, statsmodels, numpy, pandas, scipy, forestplot, and pymer4. Our NLP models are available on the Hugging Face hub at <https://huggingface.co/CNT-UPenn>, and our code is available on GitHub at https://github.com/penn-cnt/NLP_Disparities_in_Seizure_Freedom. We do not make our data available to protect patient privacy.

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